Transport in animals

Mark scheme and worked solutions for 30 questions: 15 multiple choice, 15 structured.

1 · Part 1

Core · Circulatory systems · [1]

Answer: B

Explanation

Valves close when blood starts to flow backwards, so blood can only move one way. Thick artery walls withstand high pressure, thin capillary walls allow exchange, and platelets are involved in clotting; none of these controls the direction of flow.

2 · Part 1

Extended · Circulatory systems · [1]

Answer: D

Explanation

In a fish, blood goes heart → gills → body → heart, passing through the heart once per circuit. In a mammal, blood goes heart → lungs → heart → body → heart, passing through the heart twice. The other rows reverse or confuse the two systems.

3 · Part 1

Core · Heart · [1]

Answer: A

Explanation

Arteries carry blood away from the heart and veins return it to the heart. Option B reverses the two, option C ignores the return route, and capillaries link arteries to veins rather than returning blood directly to the heart.

4 · Part 1

Core · Heart · [1]

Answer: C

Explanation

The activity of the heart can be monitored by an ECG, by measuring the pulse rate and by listening to the sounds of the valves closing. Breathing rate and body temperature do not directly monitor the heart.

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5 · Part 1 Core Heart [1]

Answer: A

Worked solution

Percentage increase = (new value − original value) ÷ original value × 100. Always divide by the original (resting) value. Check: 72 × 1.75 = 126.

6 · Part 1 Core Heart [1]

Answer: B

Explanation

Coronary heart disease is the blockage of the coronary arteries, which supply the heart muscle itself, so the muscle receives less oxygen and glucose. Blockage of the vena cava or pulmonary artery, or damaged valves, are not coronary heart disease.

7 · Part 1 Extended Heart [1]

Answer: D

Explanation

The left ventricle pumps blood through the aorta to the whole body, so it must produce a higher pressure and needs more muscle. Both ventricles pump the same volume per beat, the right ventricle (not the left) pumps to the lungs at low pressure, and the vena cava enters the right atrium.

8 · Part 1 Extended Heart [1]

Answer: C

Explanation

When the ventricles contract, the atrioventricular valves close to stop blood flowing back into the atria, and the semilunar valves open so blood can leave through the arteries. Row A would allow backflow into the atria; rows B and D would stop blood leaving the heart.

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9 · Part 1 Core Blood vessels [1]

Answer: B

Explanation

Veins carry blood at low pressure, so they have thin walls, a wide lumen and valves to prevent backflow. Row A describes an artery, row C lacks the valves found in veins, and row D has a thick wall, which is a feature of arteries.

10 · Part 1 Core Blood vessels [1]

Answer: C

Explanation

In capillaries, oxygen and glucose diffuse from the blood into the respiring muscle cells, and carbon dioxide diffuses from the cells into the blood. Options A and B reverse the direction of movement, and red blood cells stay inside the capillaries.

11 · Part 1 Extended Blood vessels [1]

Answer: D

Explanation

Blood in veins is at low pressure, so it could flow backwards; valves close to prevent this. Blood in veins is not at high pressure, valves do not strengthen the wall, and valves do not contract or pump.

12 · Part 1 Core Blood [1]

Answer: A

Explanation

Plasma transports dissolved substances, including urea, hormones and carbon dioxide. Platelets are involved in clotting, red blood cells transport oxygen, and white blood cells defend against pathogens.

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13 · Part 1 Core Blood [1]

Answer: D

Explanation

Red blood cells contain haemoglobin, which combines with oxygen in the lungs and releases it in the tissues. Phagocytosis is carried out by white blood cells, red blood cells transport oxygen rather than using it for other cells, and antibodies are made by white blood cells and do not carry oxygen.

14 · Part 1 Extended Blood [1]

Answer: B

Explanation

Lymphocytes produce antibodies and phagocytes engulf pathogens by phagocytosis. Row A swaps the functions, row C gives phagocytes the wrong function, and clotting involves platelets, not lymphocytes.

15 · Part 1 Extended Blood [1]

Answer: C

Explanation

During clotting, fibrinogen is converted into fibrin, which forms a mesh that traps blood cells to make a clot. Option A reverses the conversion, platelets are not converted into fibrinogen, and fibrin does not digest pathogens.

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16 · Part 1 Core Circulatory systems [4]

Mark scheme

(a) **[1]**

- heart (1)

(b) **[1]**

- to ensure one-way flow of blood / to prevent backflow of blood (1)

(c) **[2]**

- some blood flows backwards (through the valve) (1)
- less blood flows forwards / less blood reaches the organs or tissues (1)
- heart has to work harder / beat faster to move the same amount of blood (1)
- less oxygen or glucose reaches the tissues (1)
- max 2

Worked solution

The heart is the pump that pushes blood through the vessels. Valves are flaps that close when blood starts to flow the wrong way, so blood can only move in one direction. If a valve cannot close fully, some blood leaks back the wrong way each time the heart beats. This means less blood moves forward to the organs, so the tissues receive less oxygen and glucose, and the heart has to work harder to keep the blood moving.

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17 · Part 1 Extended Circulatory systems [6]

Mark scheme

(a) **[2]**

- blood is pumped from the heart to the gills (1)
- (then flows from the gills) to the rest of the body and back to the heart / passes through the heart once in each circuit (1)

(b) **[3]**

- one circuit from the heart to the lungs and back to the heart (1)
- a second circuit from the heart to the rest of the body and back to the heart (1)
- blood passes through the heart twice in each complete circuit (1)

ALLOW pulmonary circulation for the lungs circuit and systemic circulation for the body circuit

(c) **[1]**

- left (side) (1)

ALLOW left atrium

Worked solution

In a fish, the heart pumps blood to the gills, where it picks up oxygen. The blood then carries on to the rest of the body and returns to the heart. It goes through the heart only once on each trip round the body, so this is a single circulation. In a mammal there are two circuits. The right side of the heart pumps blood to the lungs, and the blood returns to the left side of the heart. The left side then pumps this oxygenated blood around the rest of the body, and it returns to the right side. Blood passes through the heart twice on each complete trip, so this is a double circulation.

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18 · Part 1 Extended Circulatory systems [6]

Mark scheme

(a) **[2]**

- (4.0 − 2.4) ÷ 4.0 × 100 / 1.6 ÷ 4.0 × 100 (1)
- 40% (1)

ALLOW correct answer with no working for 2 marks

(b) **[4]**

- (in a mammal) blood returns to the heart after the lungs and is pumped again (1)
- so blood to the body is at a higher pressure / quote 12.0 kPa (in the aorta) compared with 2.4 kPa (to the fish's body) (1)
- so blood flows faster to the tissues / oxygen and glucose are delivered more quickly (1)
- supports a higher rate of respiration / higher metabolic rate / maintaining body temperature (1)
- low pressure to the lungs / quote 2.0 kPa in the pulmonary artery, prevents damage to the lungs (capillaries) / gives more time for gas exchange (1)
- oxygenated and deoxygenated blood are kept separate (1)
- max 4

Worked solution

(a) The pressure falls from 4.0 kPa to 2.4 kPa, a decrease of 1.6 kPa. Percentage decrease = 1.6 ÷ 4.0 × 100 = 40%. Check: 40% of 4.0 is 1.6, and 4.0 − 1.6 = 2.4.

(b) In a fish, blood loses a lot of pressure in the narrow vessels of the gills, and it goes straight on to the body at only 2.4 kPa. In a mammal, blood returns to the heart after the lungs and is pumped a second time, so blood leaves in the aorta at 12.0 kPa. This higher pressure moves blood quickly to the tissues, so oxygen and glucose arrive fast enough for a high rate of respiration, which mammals need to stay active and keep their body temperature up. At the same time, the separate circuit to the lungs is at low pressure (2.0 kPa), which protects the delicate lung capillaries and gives time for gas exchange.

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19 · Part 1 Core Heart [6]

Mark scheme

(a) **[1]**

- place fingers on the wrist / neck and count the number of beats in one minute (1)

ALLOW count for 15 s and multiply by 4 / count for 30 s and multiply by 2

ALLOW use a (digital) pulse or heart rate monitor

(b) **[1]**

- 60 (beats per minute) (1)

(c) **[2]**

- pulse rate decreases (after exercise stops) (1)
- decreases quickly at first and then more slowly / levels off at 74 beats per minute (from 5 minutes) (1)
- quotes data, e.g. falls from 132 to 101 in 2 minutes, or still higher than the resting rate of 72 after 6 minutes (1)
- max 2

(d) **[2]**

- type of exercise / height of the bench / rate of stepping (1)
- length of time exercising / 3 minutes (1)
- length of rest before the first measurement (1)
- method of measuring pulse / length of time beats are counted (1)
- max 2

Worked solution

(a) Press two fingers gently on the inside of the wrist or the side of the neck and count the beats for one minute (or for 15 s and multiply by 4).

(b) Increase = 132 − 72 = 60 beats per minute. Check: 72 + 60 = 132.

(c) After exercise the pulse rate falls. It falls quickly at first, from 132 to 101 beats per minute in the first 2 minutes, and then more slowly. By 5 minutes it has levelled off at 74 beats per minute, which is still slightly above the resting rate of 72.

(d) For a fair comparison, both people must do the same exercise in the same way, for example the same bench height, the same stepping rate and the same 3 minutes, and measure their pulses in the same way.

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20 · Part 1 Core Heart [8]

Mark scheme

(a) **[2]**

- the coronary arteries become blocked / narrowed (by fatty deposits) (1)
- less oxygen / glucose / blood reaches the heart muscle (1)
- heart muscle cells cannot respire (aerobically) and may die / heart attack (1)
- max 2

(b) **[2]**

- any two from: genetic predisposition / family history, age, sex (2)

REJECT diet, lack of exercise, stress, smoking

(c) **[4]**

- (diet) eat less saturated fat / cholesterol (1)
- so fewer fatty deposits build up in the coronary arteries (1)
- (diet) eat less salt / avoid excess energy intake, so less risk of high blood pressure / obesity (1)
- (exercise) strengthens the heart muscle / lowers blood pressure (1)
- (exercise) helps to reduce body mass / prevents obesity (1)
- diet and exercise reduce the risk but cannot remove it, because some risk factors cannot be changed (1)
- max 4, at least one point about diet and one about exercise for full marks

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Worked solution

(a) The coronary arteries supply the heart muscle with blood. In CHD, fatty deposits narrow or block these arteries, so less oxygen and glucose reach the heart muscle. Without enough oxygen, the muscle cells cannot respire properly and may die, causing a heart attack.

(b) Genetic predisposition, age and sex are risk factors that cannot be changed. Diet, exercise, stress and smoking can all be changed.

(c) Eating less saturated fat reduces the build-up of fatty deposits in the coronary arteries. Eating less salt and not eating more energy than you use help to prevent high blood pressure and obesity. Regular exercise strengthens the heart muscle, lowers blood pressure and helps to keep body mass healthy. These changes lower the risk of CHD, but they cannot remove it completely, because factors such as age and genes still apply.

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21 · Part 1 Extended Heart [6]

Mark scheme

(a) **[1]**

- 3.5 : 1 (1)

(b) **[3]**

- left ventricle pumps blood (through the aorta) to the whole body / rest of the body (1)
- right ventricle pumps blood only to the lungs / over a shorter distance (1)
- left ventricle must produce a higher pressure (so needs more muscle) (1)
- lungs need low pressure to avoid damaging the (lung) capillaries (1)
- max 3

(c) **[2]**

- atria only pump blood into the ventricles / over a short distance (1)
- so only a low pressure is needed / ventricles must pump blood out of the heart at higher pressure (1)

Worked solution

(a) Ratio = 14 ÷ 4 = 3.5, so the ratio is 3.5 : 1. Check: 4 × 3.5 = 14.

(b) The left ventricle pumps blood through the aorta to the whole body, which is a long way and needs a high pressure. The right ventricle only pumps blood to the nearby lungs. The lungs need a lower pressure, because high pressure would damage their delicate capillaries. More muscle in the left ventricle wall gives a more powerful contraction and a higher pressure.

(c) The atria only push blood a short distance, down into the ventricles, so they need little force. The ventricles must push blood out of the heart into the arteries, so they need much thicker muscle walls.

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22 · Part 1 Extended Heart [6]

Mark scheme

(a) **[5]**

- muscles of the (right) atrium contract (1)
- (blood pushed) through the (open) atrioventricular valve into the (right) ventricle (1)
- muscles of the (right) ventricle contract (1)
- atrioventricular valve closes to prevent backflow into the atrium (1)
- semilunar valve opens and blood is pushed into the pulmonary artery (1)
- semilunar valve closes (when the ventricle relaxes) to prevent backflow from the pulmonary artery into the ventricle (1)
- max 5

ALLOW tricuspid valve for atrioventricular valve

(b) **[1]**

- (the) valves closing (1)

Worked solution

(a) Blood collects in the right atrium. The muscle in the atrium wall contracts and pushes the blood through the open atrioventricular valve into the right ventricle. The thick muscle of the ventricle wall then contracts. The rising pressure closes the atrioventricular valve, so blood cannot flow back into the atrium, and opens the semilunar valve, so blood is pushed out into the pulmonary artery. When the ventricle relaxes, the semilunar valve closes, which stops blood flowing back from the pulmonary artery into the ventricle.

(b) The sounds of the heartbeat are made by the valves closing.

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23 · Part 1 Extended Heart [6]

Mark scheme

(a) **[4]**

- muscles contract more / faster during more intense activity (1)
- so they need more energy (1)
- so the rate of (aerobic) respiration in muscle cells increases (1)
- more oxygen and glucose must be delivered to the muscles (1)
- more carbon dioxide must be removed (from the muscles) (1)
- heart beats faster to increase the rate of blood flow (to the muscles) (1)
- max 4

(b) **[2]**

- lactic acid built up (in muscles and blood) during (vigorous) exercise / oxygen debt (1)
- (fast heart rate) transports lactic acid in the blood from the muscles to the liver (1)
- (fast heart rate) transports oxygen for aerobic respiration of lactic acid (in the liver) (1)
- max 2

Worked solution

(a) The harder the cyclist works, the faster her leg muscles contract and the more energy they need. This energy comes from respiration, so the muscle cells respire faster and need more oxygen and glucose. They also produce more carbon dioxide, which must be removed. The heart beats faster, rising from 64 to 182 beats per minute, so that blood flows to the muscles more quickly to deliver these substances and remove carbon dioxide.

(b) During the sprint the muscles could not get enough oxygen, so some anaerobic respiration happened and lactic acid built up, causing an oxygen debt. After stopping, the heart keeps beating fast to carry the lactic acid in the blood to the liver, and to supply oxygen so the lactic acid can be broken down by aerobic respiration.

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24 · Part 1 Extended Heart [6]

Mark scheme

(a) **[1]**

- separates oxygenated blood from deoxygenated blood / separates the left and right sides of the heart (1)

(b) **[3]**

- oxygenated blood mixes with deoxygenated blood (1)
- blood pumped to the body (through the aorta) contains less oxygen / is not fully oxygenated (1)
- less oxygen reaches the body cells for (aerobic) respiration (1)
- so less energy is released (so the baby tires easily) (1)

> **ALLOW** some oxygenated blood passes into the right ventricle and is pumped back to the lungs, so less oxygenated blood is pumped to the body

- max 3

(c) **[2]**

- any two from: ECG, measuring pulse rate, listening to the sounds of the valves closing (2)

> **ALLOW** listening with a stethoscope for listening to the sounds of the valves

Worked solution

(a) The septum is the wall between the left and right sides of the heart. It keeps oxygenated blood on the left separate from deoxygenated blood on the right.

(b) If there is a hole, oxygenated and deoxygenated blood mix. The blood the left ventricle pumps into the aorta is then not fully oxygenated, so less oxygen reaches the body cells. The cells cannot carry out as much aerobic respiration, so less energy is released and the baby tires easily.

(c) A doctor can use an ECG, measure the pulse rate, or listen with a stethoscope to the sounds of the valves closing.

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25 · Part 1 Core Blood vessels [6]

Mark scheme

(a) **[2]**

- X artery, Y vein, Z capillary (2)
- one or two correct (1)

(b) **[1]**

- (Y) has valves (1)

> **ALLOW** thinner wall than X / wider lumen than X

(c) **[1]**

- 625 (1)

(d) **[2]**

- exchange of substances between the blood and the tissues / cells (1)

> **ALLOW** named substances, e.g. oxygen and glucose to cells, carbon dioxide from cells

- carry blood through the tissues / link arteries to veins / link arterioles to venules (1)

> **ALLOW** supply of oxygen and glucose to cells and removal of carbon dioxide and urea from cells, credited as two separate functions

- max 2

Worked solution

(a) X has the thickest wall and a narrow lumen with no valves, so it is an artery. Y has a thinner wall, a wide lumen and valves, so it is a vein. Z is microscopic, with a wall only 0.001 mm thick, so it is a capillary.

(b) Only veins have valves, so the valves in Y show it is a vein.

(c) 5.0 ÷ 0.008 = 625, so the lumen of Y is 625 times wider than the lumen of Z. Check: 0.008 × 625 = 5.0.

(d) Capillaries are where substances such as oxygen, glucose and carbon dioxide are exchanged between the blood and the body cells. They also link the arteries to the veins, carrying blood through the tissues.

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26 · Part 1 Extended Blood vessels [6]

Mark scheme

(a) **[3]**

- thick wall (1)
- to withstand the high pressure / so the wall does not burst (1)
- (wall contains) elastic tissue / elastic fibres (1)
- which stretches and recoils with each heartbeat / as pressure rises and falls (1)
- (wall contains) muscle (1)
- max 3

(b) **[3]**

- thin wall, because the pressure is low (1)
- wide lumen (1)
- so there is less resistance to blood flow (at low pressure) (1)
- valves to prevent backflow (because pressure is low) (1)
- max 3

Worked solution

(a) The artery receives blood straight from the heart at high pressure, which surges with each heartbeat. Its thick wall withstands this pressure without bursting. The wall contains elastic tissue, which stretches as each surge arrives and then recoils, helping to push the blood along and smooth out the flow. It also contains muscle.

(b) Blood in the vein is at low pressure, so a thick wall is not needed. The wide lumen gives less resistance, so blood can still flow back to the heart at low pressure. Because the pressure is so low, blood could flow backwards, so veins have valves that close to prevent backflow.

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27 · Part 1 Extended Blood vessels [6]

Mark scheme

(a) **[2]**

- (wall one cell thick) gives a short diffusion distance (1)
- so oxygen / glucose diffuse quickly from the blood into the cells / carbon dioxide diffuses quickly from the cells into the blood (1)

(b) **[2]**

- red blood cells are pushed close to the capillary wall / close to the tissue cells (1)
- so the diffusion distance for oxygen is short (1)
- blood flows slowly (through the narrow lumen), so there is more time for exchange (1)
- max 2

(c) **[2]**

- large surface area for exchange / diffusion (1)
- every cell is close to a capillary / short diffusion distance to each cell (1)
- cells receive enough oxygen and glucose (for respiration) and have carbon dioxide removed (1)
- max 2

Worked solution

(a) Because the wall is only one cell thick, substances have only a very short distance to travel between the blood and the tissue fluid around the cells. Oxygen and glucose therefore diffuse out of the blood quickly, and carbon dioxide diffuses in quickly.

(b) In a narrow capillary the red blood cells squeeze through one at a time and are pushed against the wall. The oxygen they carry has only a short distance to diffuse into the tissues. Blood also moves slowly through capillaries, giving more time for exchange.

(c) A large, branched network gives a very large surface area for exchange, and it means no cell is far from a capillary. Every cell can therefore get the oxygen and glucose it needs for respiration and have its carbon dioxide removed quickly.

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28 · Part 1 Core Blood [8]

Mark scheme

(a) **[2]**

- red blood cells, white blood cells, platelets, plasma: all four (2)
- two or three correct (1)

(b) **[1]**

- (involved in) blood clotting (1)

(c) **[2]**

- phagocytosis / engulfing pathogens (1)
- producing antibodies (1)

(d) **[1]**

- combines with / carries oxygen (1)

ALLOW picks up oxygen in the lungs and releases it in the tissues

(e) **[2]**

- any two from: carbon dioxide, hormones, nutrients / named nutrient (e.g. glucose, amino acids), ions / named ion, blood cells (2)

REJECT urea (named in the question)

REJECT oxygen

Worked solution

(a) Blood is made of red blood cells, white blood cells, platelets and plasma.

(b) Platelets are involved in blood clotting.

(c) White blood cells defend the body against pathogens. Some engulf pathogens by phagocytosis and others produce antibodies.

(d) Haemoglobin combines with oxygen in the lungs and releases it in the body tissues, so red blood cells can transport oxygen.

(e) Plasma is the liquid part of blood. It transports blood cells, ions, nutrients such as glucose, urea, hormones and carbon dioxide.

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29 · Part 1 Core Blood [6]

Mark scheme

(a)(i) **[2]**

- prevents (further) blood loss (1)
- prevents entry of pathogens (1)

(a)(ii) **[1]**

- platelets (1)

(b) **[3]**

- less haemoglobin (in the blood) (1)
- so less oxygen is transported to the body cells / tissues (1)
- less aerobic respiration (in cells) (1)
- so less energy is released (1)
- max 3

Worked solution

(a) When the clot forms it seals the cut. This stops more blood being lost and stops pathogens such as bacteria getting into the body. Platelets are the part of the blood involved in clotting.

(b) Red blood cells contain haemoglobin, which carries oxygen. With fewer red blood cells, less oxygen reaches the body cells. The cells cannot carry out as much aerobic respiration, so they release less energy, and the person feels tired.

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30 · Part 1 Extended Blood [6]

Mark scheme

(a) **[2]**

- fibrinogen is converted to fibrin (1)
- fibrin forms a mesh (that traps blood cells to form a clot) (1)

(b)(i) **[1]**

- engulf pathogens / bacteria by phagocytosis (1)

(b)(ii) **[1]**

- produce antibodies (1)

(c) **[2]**

- number increases (from day 0 to day 2) because phagocytes move to the site of infection / respond to the bacteria (1)
- number decreases (after day 2) because the bacteria have been destroyed / engulfed, so fewer phagocytes are needed (1)

ALLOW phagocytes die after engulfing bacteria

REJECT phagocytes produce antibodies

Worked solution

(a) At the cut, the soluble protein fibrinogen in the blood is converted into fibrin. Fibrin forms a mesh of fibres that traps blood cells and forms a clot.

(b) Phagocytes engulf and digest pathogens by phagocytosis. Lymphocytes produce antibodies.

(c) The number of phagocytes rises from 2 to 34 by day 2, because more phagocytes move to the site where the bacteria are, in order to engulf them. After day 2 the number falls, to 9 by day 5, because most of the bacteria have been engulfed and destroyed, so fewer phagocytes are needed.

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